Loup Garou: Game theory and optimal strategy

Analysis of the Werewolf game

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Agenda

What is Loup Garou?

Basic setup (N players):

- X Civilians (including special roles)
- Y Werewolves
- 2 phases per cycle

Win conditions:

- Villagers win: All werewolves eliminated
- Werewolves win: Equal or outnumber villagers

Key asymmetry

Werewolves have **complete information** about each other, while villagers operate under **incomplete information**.

Game flow

Night phase:

- Werewolves secretly coordinate and eliminate one villager
- Special roles (Seer, Witch) may act

Day phase:

- All players vote publicly to eliminate one suspect
- Goal: Identify and eliminate werewolves
- Deduction and rhetoric determine outcomes

Information asymmetry

Werewolves vote as a **coordinated bloc** while villagers vote **individually with partial information**.

Bayesian game framework

Formal definition:

- Players (considering a 13 player setup): $N = \{1, 2, \ldots, 13\}$
- Types: $\Theta = \{\text{Werewolf}, \text{Villager}\}$
- Actions: $A_i = \{\text{Vote for player } j : j \neq i\}$
- Information: Players have **asymmetric beliefs** about types

Payoff functions:

$$u_i(\text{action profile}, \text{true types}) = \begin{cases} 1 & \text{if player } i\text{'s faction wins} \\ 0 & \text{otherwise} \end{cases}$$

Incomplete information framework

Unlike chess (perfect information), players cannot compute optimal actions without modeling uncertainty over others' types.

Perfect Bayesian Equilibrium (PBE)

A PBE is a strategy profile where:

1. Each player's strategy is optimal **given** their beliefs about others' types
2. Beliefs are updated using Bayes' rule when possible
3. At information sets reached with zero prior probability, beliefs can be arbitrary

Why it matters for Loup Garou:

- Werewolves must maintain cover while coordinating (belief constraint)
- Villagers must infer types from voting patterns (Bayesian updating)
- *"Lying as an equilibrium"*: False accusations can be rational if undetected

Situational example

A werewolf voting to eliminate another werewolf (seemingly random) can confuse villagers, preventing pattern recognition.

The "Random Strategy" framework

Academic Breakthrough (Wang, 2025):
Standard models assumed *random voting* was optimal. Wang proved that coordinated **random selection** is more profitable.

Mechanism:

1. Each player announces natural number $x_1, x_2, \ldots, x_{13}$
2. Target player: $t = (\sum x_i \bmod 13)$
3. Any deviator is immediately exposed as werewolf

Key insight:

- Werewolves can **coordinate** within the rule structure
- Villagers cannot detect coordination because voting appears random
- This is a **Perfect Bayesian Equilibrium**

Strategic advantage

Enforcement mechanism prevents deviation $\rightarrow$ Wolves can trust each other's voting while maintaining cover.

The "All-In Strategy" (critical phase)

When parity approaches: $\#\text{Werewolves} = \#\text{Villagers}$

Scenario 1: Modulo targets villager

- Wolves follow the rule
- Next night: $\#W = \#V = 2$
- Wolves win (controlling vote)

Scenario 2: Modulo targets wolf

- All wolves vote for same villager
- Forcing a tie
- 50% win on random resolution

Outcome

Parity is mathematically equivalent to wolf victory with "all-in strategy".

Winning probability formula

Recursive formula for werewolf win rate:

$$w(n,m) = \begin{cases} 0 & \text{if } m = 0 \\ 7/8 & \text{if } n = 5, m = 2 \text{ (base case)} \\ 1 & \text{if } m \geq n - m \\ \frac{n-1-m}{n-1} w(n-2, m) + \frac{m}{n-1} w(n-2, m-1) & \text{otherwise} \end{cases}$$

where:

- $n =$ total players
- $m =$ number of werewolves

For our case ($n = 13, m = 3$):

$w(13,3) \approx 0.827$ (approximately 41-43% after accounting for transitions)

Interpretation

Despite being outnumbered 10:3, optimally-playing werewolves win *more than one out of every three games*.

Impact of parity

Counterintuitive finding:
For a fixed number of werewolves m, increasing player count sometimes *helps* wolves:

$$w(2k, m) > w(2k-1, m)$$

- Adding one villager (making n even) means **fewer voting cycles**
- Per-cycle: lower elimination probability for wolves
- Result: **odd parity favors villagers** (more voting rounds)

13 vs. 12 Players

13 players with 3 wolves is actually *worse for villagers* than 12 players with 3 wolves because odd n means you never reach a (n, m) where wolves control voting completely.

Design implication

Game organizers can design the game for **even** player counts to favor villagers!

Villager strategy with a Seer

Special role: Seer (Can verify one player per night)

Optimal revelation timing (from Monte Carlo):

Villagers	Werewolves	Optimal Round	Win %
10	3	Round 4	40%
10	2	Round 4	58%
10	1	Round 3	79%

Key insights:

1. **Reveal early** if seer verifies a wolf (credibility matters)
2. **Delay if only villagers checked** (builds information advantage)
3. **Round 3-4** is critical for 3-wolf games
4. Wolves will target seer immediately after verification

Expected payoff

Seer dramatically improves villager win rate: 27% (no seer) → 40% (with seer)

Why Monte Carlo for Loup Garou?

Traditional game tree:

- 13 players choose targets
- 13! possible vote permutations
- Day 1: 13 possibilities
- Day 2: 11 possibilities
- Exponential explosion!

Monte Carlo approach:

- Sample random game outcomes
- Apply strategic rules to samples
- Aggregate statistics
- Tractable!

Computational advantage

Instead of evaluating $13! \times 11! \times 9! \times \ldots$ outcomes, simulate $N = 10,000$ games and estimate win rates with error $O(1/\sqrt{N})$.

Monte Carlo framework

Algorithm:
Setup: Initialize game with 10 villagers, 3 werewolves
Loop (for each of 10,000 simulations):

1. Day phase: Players vote according to strategy
2. Eliminate target
3. Check win condition
4. Night phase: Werewolves eliminate (with special roles)
5. Check win condition
6. Advance day

Aggregate: Count villager wins, werewolf wins
Confidence interval: Compute 95% CI around estimated win rate
Expected runtime: ∼5-10 seconds for 10,000 simulations on standard hardware

Voter models for simulation

Key implementation question: How do players vote?

Model 1: Random voting

- Villagers: Uniform random choice
- Werewolves: Coordinated (all vote same non-wolf)
- Baseline for comparison

Model 2: Information responsive

- Seer votes for **verified werewolf** if known
- Others use Bayesian inference: *"Who's acting suspicious?"*
- Requires modeling **belief dynamics**

Model 3: Optimal with coordination

- Werewolves use modulo mechanism (Wang strategy)
- Villagers try to detect coordination patterns
- Most sophisticated, hardest to implement

Recommendation

Start with **Model 1** (baseline), then upgrade to **Model 2** for more realism.

Key simulation parameters

Parameter	Suggested value
Number of simulations	10,000
Player count	13
Werewolf count	3
Villager strategy	Random / Seer-informed
Werewolf strategy	Coordinated elimination
Seer presence	Yes / No (compare)
Witch presence	Yes / No (compare)

Analysis outputs:

- Werewolf win rate $\pm$ 95% CI
- Mean game length (days)
- Seer vs. no-Seer comparison
- Sensitivity to voter model

For werewolves: Early game

Rounds 1-3 (days 1-3):
Objective: Maintain cover while eliminating high-value villagers
Tactics:

- Avoid voting patterns: Don't always vote as a bloc
- Target high-status players: Experts, analysts, obvious leaders
- Throw suspicious votes: Vote for different villagers to appear uninformed
- Encourage chaos: Support wild accusations to muddy the waters

Specific example

Three wolves: Wolf A votes for Person 1, Wolf B votes for Person 2, Wolf C votes for Person 3. On night, all target Person 4. This creates appearance of uncoordination.

Critical Don't

Do **NOT** obviously protect each other in votes. This is the primary detection method for villagers.

For werewolves: mid game strat

Rounds 4-6 (days 4-6):
Objective: Identify and eliminate Seer / special roles
Tactics:

- Watch for *too-confident* accusations (likely Seer checking)
- Target anyone claiming to have checked you
- Use sabotage strategically (if quest mechanics present)
- Begin subtle coordination for parity approach

Mathematical consideration: After 3 day eliminations, if 3 wolves remain with 7 villagers:

$$\text{Wolf win probability} \approx 0.65 \text{ (if cover maintained)}$$

Parity countdown

Target: Reach parity (3W, 3V) by **Day 5-6**. From there, "all-in strategy" secures victory.

For werewolves: late game parity

When #Wolves $\approx$ #Villagers:
Control voting: With proper coordination, wolves now control plurality
Force final vote: Last elimination determines game outcome
All-in strategy: Pre-coordinated unanimous vote if needed
Specific scenario: 3 Wolves, 3 Villagers

- Wolves vote: Villager A, Villager A, Villager A
- Villagers vote: Wolf X, Wolf X, Wolf X
- **Result**: 3-3 tie on villager elimination
- Random tiebreaker: 50% wolf victory
- If villager eliminated: Wolves win (3W, 2V)

Endgame guarantee

Reaching parity with intact coordination essentially guarantees wolf victory.

For villagers: Anti werewolf tactics

Pattern recognition:

- Track voting patterns across days
- Statistical test: Do any 2-3 players have correlated votes?
- Bayesian update: Players with similar votes are likely same faction

Seer coordination:

- Reveal Round 4 with **verified wolf**
- Use Seer check votes to build coalition
- Protect Seer on subsequent nights (if Witch available)

Voting discipline:

- Form sub-coalitions ($n - 2$ players) voting as bloc
- Werewolves cannot outvote disciplined group
- Increases elimination probability from $1/13$ to $1/5$ per round

Coalition math

If 8 villagers vote as bloc: Their target gets ≥ 8 votes. Werewolves have only 3 votes. **Coordination solves the game for villagers**.

For Villagers: Seer strategy (detailed)

Night 1-2: Check "suspicious" players

- Target: High-status players, eloquent speakers, confident accusers
- Werewolves often adopt leadership roles

Day 3-4: Analyze results

- If checked player is wolf: Prepare revelation
- If checked player is villager: **Stay hidden**, check again

Day 4: Revelation decision

- Reveal 1-2 verified wolves → Build coalition to eliminate them
- Risk: Werewolves target Seer immediately
- Mitigation: Pair with Witch (resurrect Seer if killed night of revelation)

Optimal equilibrium

With coordinated seer + witch + villager coalition, villager win probability jumps to $\geq 40\%$ from 27%.

Key takeaways

Theory: Werewolves have 41-43% win rate under optimal "random strategy+" despite 3:10 disadvantage
Parity: The "all-in strategy" at parity guarantees wolf victory with proper coordination
Seer impact: Special roles dramatically shift probabilities (27% wolf wins with Seer vs. 43% without)
Monte Carlo: Simulation enables rapid analysis of complex game variants
Practice: Villager coalition-voting is the strongest anti-wolf tactic; wolf coordination is key to their success
Game design: Mechanics like "salvation assembly" can significantly alter equilibria

Final thought

Loup Garou combines **incomplete information** (Bayesian), **coordination** (game theory), and **deception** (psychology). Monte Carlo simulation captures this complexity in a way pure analysis cannot.